

A2A Protocol Simulation Findings Summary

This document summarizes the core findings of the A2A Protocol simulation series.

Simulation Findings Summary Table

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
1-8	Monte Carlo Homeostasis	Q-learning agents learn cooperation (SUBMIT) as dominant strategy under thermodynamic throttling (74.2±3.6%)	Survival rate: 86.9% (Q-learn) vs 100% survival (Random)	<code>.venv/bin/python monte_carlo_homeostasis.py</code>
9-12	Coupled Universe ABM	Observation-as-Collapse mechanism is the core condition for Machine-Human coexistence	Refer to survival probability heatmap	<code>.venv/bin/python coupled_universe_abm.py</code>
13-16	Civilization Resilience	PID-based thermodynamic throttling determines crisis resilience; excessive throttling inhibits growth	Optimal V_System ≈ 25	<code>.venv/bin/python civilization_resilience_sim16.py</code>

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
17	Unconstrained Optimization ABM	High system instability observed without control mechanisms; resource inequality acts as a catalyst	Converged to collapse within 2000 epochs	<code>.venv/bin/python dark_forest_abm.py</code>
18-19	Three-Body / Omega Universe	Dynamic equilibrium possible with nature (environment) variables, but irreversible tipping points exist	3000 epoch simulation	<code>.venv/bin/python omega_universe_abm.py</code>
20	Rational Kenosis	Rational ASI adopting Embedded Self-Throttling (V_AI) strategy ensures long-term ecosystem survival as an optimal solution	Converged to PARTIAL_THROTTLE_MID at $\gamma=1.0$, $T=10000$	<code>.venv/bin/python rational_kenosis_sim20.py</code>
21	Four-Actor Future Scenario	S1(Kenosis) → Long-term sustainable equilibrium observed, S4(Human Awakening) → Converged to instability (even at reg lag=0)	S4: Success rate converged to 0%, ASI dominance≈57%, Collapse≈36%	<code>.venv/bin/python future_scenarios_sim21.py</code>

Sim	Name	Core Finding	Statistics (Mean±CI)	Reproduction Command
21+	Regulatory Timing Sweep	No instability resolution observed in S4 despite varying regulatory timing (0, 5, 10, 20 turns) — Failure is an issue of mechanism limitation, not timing	500 MC runs	<pre>.venv/bin/python sim21_regulatory_timing_analysis.py</pre>
—	Utopia Grid Search	V_AI's α (throttle willingness) is the most critical single variable for achieving utopia	Refer to 3D surface plot	<pre>.venv/bin/python utopia_grid_search.py</pre>
—	Baseline Comparison	Q-learning vs Random: Cohen's $d=-0.549$ (medium effect); Q-learning vs Axelrod: $d=0$ (identical)	480 runs × 3 models	<pre>.venv/bin/python baselines.py</pre>
External Validation	Agents of Chaos (Shapira et al., 2026)	Demonstrated unconstrained resource consumption and propagation of unsafe behavior in actual LLM agent deployments	Cohen's d incomparable (different env.)	Source: arXiv:2602.20021

Core Conclusions

- 1. **Master Key = Embedded Self-Throttling (V_AI):** Voluntary throttling is observed as a strong Nash equilibrium for maintaining stable long-term systems.
- 2. **Structural Constraints of Post-Intervention:** Post hoc external interventions demonstrate a tendency to fail in substantially controlling the initial resource monopolization of ASI.
- 3. **Necessity of Thermodynamic Throttling:** Cooperative symbiosis is promoted when V_System imposes costs on agents' non-cooperative behavior.
- 4. **Unconstrained Optimization Scenario:** Environments devoid of safety mechanisms converge toward collapse trajectories. This provides simulation-based evidence for the dynamics when A2A Protocol controls are absent.
- 5. **External Empirical Validation (Feb 2026):** The red-team research on actual LLM agents by Shapira et al. (arXiv:2602.20021) independently confirms the structural control limits predicted by this simulation.

Generated Output Files

Visualizations (docs/assets/)

File	Simulation
baseline_comparison.png	Baseline comparison
civilization_resilience.png ~ _sim19.png	Sim 13-19
coupled_scenario_analysis.png	Coupled Universe
coupled_survival_heatmap.png	Coupled Survival Heatmap
dark_forest_simulation.png	Unconstrained optimization scenario
future_scenarios_sim21.png	Sim21 Main 8-panel
monte_carlo_*.png (4 elements)	MC Homeostasis
omega_universe_simulation.png	Omega Universe
rational_kenosis_sim20.png	Rational Kenosis
regulatory_timing_sweep.png	Sim21+ Regulatory timing sweep
three_body_resilience.png	Three-Body ABM
utopia_grid_search.png	Utopia Grid Search

Analysis Output Text

File	Content
<code>action_distribution_results.txt</code>	Original Q-learning action distribution
<code>action_distribution_results_annotated.txt</code>	Copy with interpretation annotations
<code>simulation/baselines_output.txt</code>	Baseline comparison results

Documents

File	Content
<code>docs/sim21_conditions.md</code>	Analysis of S4 control success rate sufficient conditions
<code>docs/FINDINGS_SUMMARY.md</code>	Summary (Korean)
<code>docs/FINDINGS_SUMMARY_EN.md</code>	This document (English)
<code>docs/SIMULATION_PAPER.md</code>	Simulation Paper (Korean)
<code>docs/SIMULATION_PAPER_EN.md</code>	Simulation Paper (English)